SkillUp

In collaboration with IBM SkillsBuild

AI Racing League · Session 1 of 4

From idea to first lap with IBM Bob

How AI changed the way we work and code, IBM Bob on everyday jobs, then your first look at the race car.



Tuesday 29 September 2026
3:00 – 4:30 PM GMT · 90 minutes · no coding experience needed

Today: AI at work first, the car second

0:00	Welcome and how the four sessions build	Talk
0:05	How AI changed the way we work and code, meet IBM Bob	Talk
0:17	Part 1 · Bob at work: brainstorm, plan, build, troubleshoot, document, automate	Follow along
0:48	The challenge: the mission, how the car thinks, Python in 60 seconds	Talk + demo
0:55	Part 2 · Car on the grid, baseline lap, Bob explains the car's code	Follow along
1:25	Before Session 2, support, questions	Wrap



24 min talk · 61 min hands-on with IBM Bob · 5 min wrap. The bar at the bottom of every slide shows where we are.

Welcome 0:00 – 0:05

Four sessions, each one builds on the last

Tue 29 Sep

AI at work, meet the car

Bob on everyday jobs. TORCS running, a baseline lap, Bob explains the car's code. No car edits yet.

Thu 1 Oct

Python for racers

Read the driver code with Bob, learn what you can tune, make your first one-change upgrade.

Tue 6 Oct

Engineer the corners

Plan mode for braking and corner speed, reading track sensors, fixing crashes with Bob.

Thu 8 Oct

Race ready

Zero-damage runs, Bob-written documentation, submission prep.

Submit by Sun 11 Oct

Projects go to SkillUp through the support email for evaluation.

Learning plan is individual

Teams are welcome for the project, but everyone completes the mandatory activities themselves.

Closing ceremony Thu 22 Oct

Top 3 receive the winner certificate. Everyone who completes receives the program certificate.

Words you'll hear today

IBM Bob	An AI partner inside VS Code that explains, plans and writes code with you	**Prompt**	What you type to Bob: your request in plain English
VS Code	A free editor for code files, like Word for programs	**Python**	The programming language that drives our race car
Script	A file of instructions, e.g. snakeoil3_gym.py	**Terminal**	A window where you type commands to run things
TORCS	The Open Racing Car Simulator, the game our code drives	**Diff**	A before-and-after view of what Bob changed

How building software changed

Before

- Every line written by hand
- A specialist needed at every stage
- Stuck? Search forums and documentation alone
- New language means weeks of learning syntax

Now

- Describe what you want in plain language
- An AI partner supports every stage
- Stuck? Ask for an explanation in plain words
- You spend your time reviewing and deciding

> AI isn't changing the steps of development. It's changing how those steps get done, and who can do them.

AI across the development lifecycle

1

Idea and discovery

Brainstorm, ask questions, see what's possible

Bob: Ask

2

Requirements and planning

Turn an idea into goals, features and steps

Bob: Plan

3

Build

Generate and update code from instructions

Bob: Code

4

Validate and improve

Find issues, suggest and apply fixes

Bob: Ask + Code

5

Document and deliver

Explain the work so others can use and maintain it

Bob: Code

Part 1 today walks through all five stages with Bob, using one bakery scenario. The racing challenge uses the same five stages across four sessions.

Not just for developers

Developer

Plan a feature, generate starter code, fix bugs, explain errors

Educator

Turn notes into a lesson outline, build a simple quiz page

Healthcare

Decide what a patient intake form needs, improve a workflow

Finance

Map a reporting workflow, build a small tracking tool

Small business

Plan and build a website, update it from customer feedback

Under the hood are large language models trained on code, like the IBM Granite models from your learning plan. They can generate, explain and fix code in over 100 languages.

Quick check from your pre-work

Anna must build a new feature for her team's website but doesn't know where to start. How should she use Bob?

A Start building in Code mode and fix it later

B Gather all requirements alone, then have Bob generate a near-complete solution

C Use Bob to understand the problem, plan, then build and refine

D Have Bob sketch a basic outline, then do the rest by hand

Meet IBM Bob

Understand

Ask

Answers questions and explains files. Changes nothing.

Decide

Plan

Turns a goal into a step-by-step plan before anything is built.

Change

Code

Creates and edits files. Shows you a diff and waits for approval.

> Keep auto-approval off. Bob proposes, you decide.

Prompts that get results

Vague

fix my website

Structured

Find all syntax and functional errors in @Menu.html. Group them by severity. Say what each one breaks. Don't edit yet.

Scope

Which file, which part

Goal

What "done" looks like

Shape

Grouped, ranked, step by step

Guardrails

"Don't edit yet", "keep the structure"

Part 1 0:17 – 0:48

Part 1 of 2

Bob at work: one bakery, eight jobs

We open Sprout and Crumb, a vegan bakery, and use Bob to brainstorm, gather requirements, plan, build, troubleshoot, document, automate and explain errors.

Follow along

Your learner sheet has every prompt

Stuck?

Just watch, it's all repeatable later

Checkpoint

A working page and your first Python run

SkillUp

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Part 1 · Brainstorm 0:17 – 0:20 · 3 min

1 · Brainstorm

Ask mode

I'm opening a small vegan bakery called Sprout and Crumb. I can't code.
Brainstorm 8 things a website could do for a bakery like mine.
Put them in a table with the columns: idea, who it helps, effort (low / medium / high).
Keep the language simple.

Skill

Exploring ideas and what's possible before committing to anything.

Stage

Idea and discovery

2 · Gather requirements

Ask mode

From your list, I want: a menu, our story, and a preorder form.
Turn these into clear requirements for the website.
Group them as must have, should have, nice to have.
For the preorder form, list every field and what makes an entry valid.
Ask me up to 3 questions about anything that's unclear.

Then · Ask

Pickup only, Tuesday to Saturday. Orders need 48 hours' notice. No payment online.

Skill

Turning ideas into clear requirements, and letting AI ask you clarifying questions.

Stage

Requirements and planning

3 · Plan

Plan mode

Using these requirements, plan a simple one-page website for Sprout and Crumb.
Give me a step-by-step build plan: the page sections in order, what goes in each, and how the preorder form should behave.
It must be one HTML file that works on phones. Don't build it yet.

Then · Plan

Make it simpler: remove anything that needs a server or a database, and explain the plan for someone who isn't a developer.

Skill

Getting a step-by-step plan you can review and change before anything is built.

Stage

Requirements and planning

4 · Build

Code mode

Build the plan as a single file called index.html with the CSS and JavaScript inside it.
Use 6 placeholder menu items with prices in pounds.
The preorder form should check the fields are filled in and show a thank-you message.
It doesn't need to send anything.
Add short comments so a beginner can follow the code.

Then · Code

Change the colours to soft greens and add a "Today's special" banner at the top.

Skill

Turning a plan into working output with plain-language instructions, and reviewing before approving.

Stage

Build

5 · Improve and troubleshoot

Code mode

Can you identify all syntax and functional errors in this HTML file: @Menu.html?
Perform a thorough analysis.
Group the findings by severity (critical, high, medium, low).
Indicate whether each issue is related to syntax, functionality, or structure.
Identify anything that could prevent the page from rendering or behaving correctly.

Then · Code

Rewrite this file to fix the identified issues.
Apply the recommendations from your previous analysis.
Preserve the original structure where possible.

Skill

Finding and classifying errors, reviewing proposed fixes, comparing before and after.

Stage

Validate and improve

Prep: Download Menu.html from the lab's GitHub page using the Raw button, then File → Save As into bob-at-work.

6 · Document and create content

Code mode

Create README.md for the Sprout and Crumb site in @index.html, written for the bakery owner, not a developer.
Explain what each section does and how to change a menu item or price.
Then create launch-post.md with a short Instagram caption and a 5-slide outline for a talk to local cafés: a title, 3 bullet points and a visual idea for each slide.

Skill

Producing documentation and ready-to-use content, then checking it against reality.

Stage

Document and deliver

Part 1 · Automate 0:40 – 0:45 · 5 min

7 · Automate, and explain an error

Code mode

Automate a weekly job for me.
1. Create preorders.csv with 15 made-up preorders: name, item, quantity, pickup date.
2. Write summarise_orders.py that reads the file and prints the total quantity per item and the number of orders per pickup date, sorted.
Use only Python's built-in libraries. Add beginner-friendly comments. Tell me the exact command to run it.

Then · Ask

I ran summarise_orders.py and got this error:
[paste the error]
Explain it in plain English and tell me how to fix it. Don't change any files.

Skill

Automating a repetitive task with a small script, running it, and getting errors explained in plain English.

Stage

Build + improve

Your turn 0:45 – 0:48 · 3 min

Your turn: one prompt from your own work

Ask mode

I work as [your role]. I regularly have to [a task that takes you time].
Suggest three ways you could help with it, from simplest to most ambitious.
For the simplest one, show me the exact prompt I should use.

Skill

Transferring the skills to your own work.

Stage

Any stage

Same way of working, faster car

Session 1 · Ask

Understand

Café: what's wrong with the page?

Car: how does the code drive?

Session 2 · Code

First upgrade

Café: fix the menu button

Car: one change, one faster lap

Session 3 · Plan

Engineer

Café: plan the next feature

Car: brake before corners

Session 4 · Document

Deliver

Café: README for the owner

Car: explain your changes, submit

Your mission

Fastest lap.
Zero damage.

Corkscrew

Road track in TORCS

1 lap

From a standing start

scr_server 1

The car your Python code drives

IBM Bob

Your race engineer for every change

Nobody steers with a keyboard. A Python script drives the car; you and Bob improve the script.

How your car thinks

Sense

Like your eyes

How fast am I going?

Where am I on the track?

How far to the edge ahead?

→

Decide

Like your brain

The Python script's rules

"Too slow? Press the throttle."

This is what you'll improve.

→

Act

Like your hands and feet

Steer

Accelerate or brake

Change gear

TORCS the game ⇄ connection on port 3001 ⇄ **snakeoil3_gym.py** the driver

Python in 60 seconds

```
# the idea behind the throttle
if speed < target_speed:
    accel = accel + 0.01
else:
    accel = accel - 0.01
```

- # starts a note for humans, Python ignores it
- **speed, accel** are names that hold values
- **if / else** choose between two actions
- **Indented lines** belong to the line above them

Simplified from the base script. Stuck on any line of code, ever? Ask Bob: "Explain this line like I've never coded."

Part 2 of 2

Car on the grid, Bob under the bonnet

Run the unmodified car, record the time to beat, then ask Bob to explain how it drives. No code changes today.

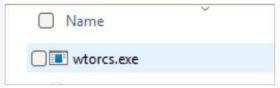
Follow along
TORCS, a terminal and VS Code

Stuck?
Post the error in chat, keep watching

Checkpoint
Baseline time in your lap log

Put the car on the grid

1. Run TORCS



2. Select Quick Race



3. Configure Race



4. Select Corkscrew Track



5. Select driver scr_server 1 (the python car)



6. Select 1 lap



7. Select New Race to start



8. Run snakeoil3_gym.py



1. Start TORCS (wtorcs.exe)
2. Race → Quick Race → Configure Race
3. Track: Road Tracks → Corkscrew
4. Driver: scr_server 1 only
5. Laps: 1 → New Race
6. Blue waiting screen means ready
7. In the terminal: cd gym_torcs, then python snakeoil3_gym.py

Part 2 0:55 – 1:25

Pit lane: quick fixes

"Connection refused" or waiting forever	TORCS must be on the blue screen first. Driver must be scr_server 1.
python not found	Reinstall Python 3 with "Add Python to PATH" ticked, then reopen the terminal.
Car spins or goes off track	Normal for the base script. Note it down; that's what we fix from Session 2.
Any other error	Paste it into Bob in Ask mode: "Explain this error and how to fix it, step by step."

Still stuck? Keep going: the next Bob steps only need the code files, not TORCS.

Part 2 0:55 – 1:25

Start your lap log

Version	What changed	Lap time	Damage	Top speed	Keep?
base	Nothing, original script	__:__.__	___	___ km/h	Yes
v1	Session 2				
v2	Session 3				

Read the time, damage and top speed from the top-left panel in TORCS when the lap ends. This log is your evidence for the final submission.

Part 2 0:55 – 1:25

Ask Bob how the car drives

Ask mode

I've never coded before. Explain how @snakeoil3_gym.py drives the race car.
1. In plain English, what does the car check and decide each moment?
2. Which values could I change to make it faster or safer? Give line numbers.
3. Why does it pick up damage in corners?
Do not change any code.

Then · Ask

Explain line [number] like I've never coded. What would happen on track if I changed it?

Skill

Using Bob to understand unfamiliar code without changing it.

Stage

Discovery

What Bob typically finds

- No braking before corners: it only reacts once it's already turning
- One fixed target speed for the whole lap
- The distance sensors aren't used to see corners coming
- Tiny throttle steps, so it's slow to get up to speed
- No racing line: it aims for the middle of the track

Team huddle, 4 min
Pick one idea you'd try first. Write it in your lap log. You'll build it with Bob in Session 2.

Wrap 1:25 – 1:30

Before Session 2 on Thursday

- [ ] Complete your mandatory learning plan activities on SkillsBuild (individually)
- [ ] Get TORCS running and record your baseline lap, if you didn't today
- [ ] Try one Bob prompt for your own day job and keep the result
- [ ] Choose your team (or go solo) and your first upgrade idea
- [ ] Post your baseline time on Discord

Wrap 1:25 – 1:30

Pit crew and links

Community and leaderboard (Zoom)

Learning plan

https://skills.yourlearning.ibm.com/activity/PLAN-6FA860C53B3A

TORCS and Python scripts

https://ibm.box.com/s/sud4g3o1mqyfqoa4ab9lr80vh8zc1ri6

Get IBM Bob

https://ibm.biz/IBMBoBUniversity

Café troubleshooting lab file

https://github.com/academic-initiative/skillsbuild/blob/main/ibm-bob/troubleshoot-your-code/file/Menu.html

Windows setup video · macOS extra steps

https://youtu.be/rRbpOUH0lFQ · https://ibm.box.com/v/MacOSExtraExt

Feedback Link and QR Code

https://survey.zohopublic.in/zs/eqCGL3



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Thank you

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